DATA STRUCTURES AND ALGORITHMS **SPRING 2026**
QUIZ 04 **TOTAL MARKS: 10**

BINARY SEARCH TREE

PROBLEMS: (Each Question carries equal marks)

1. In a Binary Search Tree, where is a new node inserted if its value is less than the current node?
- A) Right subtree
- B) Left subtree
- C) Root of the tree
- D) It replaces the current node

2. Which traversal of a BST produces keys in sorted (ascending) order?
- A) Pre-order
- B) Post-order
- C) In-order
- D) Level-order

3. When deleting a node with TWO children from a BST, which node is typically used as its replacement?
- A) The root node
- B) The leftmost node of the left subtree
- C) The in-order successor or the predecessor
- D) The rightmost node of the right subtree

4. To find the MAXIMUM key in a BST, which direction do you traverse?
- A) Keep going left until a nullptr is reached
- B) Keep going right until a nullptr is reached
- C) Visit the root node only
- D) Perform an in-order traversal and return the first element

5. Consider a BST where nodes 50, 30, 70, 20, 40 are inserted in that order. What is the output of an in-order traversal?
- A) 50 30 70 20 40
- B) 20 30 40 50 70
- C) 70 50 40 30 20
- D) 50 70 30 40 20

6. When deleting a LEAF node from a BST, what is the correct action?
- A) Set the node's data to 0 and leave its pointers unchanged
- B) Replace it with its left child
- C) Simply remove the node and set its parent's pointer to nullptr
- D) Replace it with the root

7. In the search function of a BST, if the search key is greater than the current node's value, what happens next?

A) The search terminates and returns false
B) The function recurses into the left subtree
C) The function recurses into the right subtree
D) The root is returned

8. Explain the three cases that must be handled when deleting a node from a BST. For each case, describe how the tree is restructured.

9. Why does an in-order traversal of a BST visit nodes in sorted order? Relate your answer to the BST property (left child < parent < right child).

10. In a delete operation on a node with one child, why is it sufficient to simply link the parent directly to that child? What would go wrong if you did not update the parent's pointer?